

MD ABU SAYEB RAYHAN

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Professional Summary

AI Engineer with 2+ years of professional experience delivering production-grade AI solutions in **Japan**. Experienced in **Computer Vision**, **Generative AI**, **Retrieval-Augmented Generation (RAG)** and end-to-end AI application development. Proven ability to optimize AI pipelines, deploy scalable solutions and deliver measurable business impact through automation, latency reduction and cost optimization.

Professional Experience

Data Scientist – Othello Financial Service Co. Ltd., Japan Oct 2024 – Mar 2026

- Built production computer vision pipelines for retail analytics using **MoveNet** and **Takumi Vision**, reduced inference latency by 70%.
- Developed a real-time NFT generation system using a **fine-tuned YOLO** (92% mAP), **MMAction2** for automated key moments extraction, reducing production costs by 50%.
- Designed and deployed an enterprise **RAG** chatbot using **LangChain**, **Pinecone**, **GPT-4o** and **Flask APIs** on Microsoft Azure for context-aware document retrieval.
- Integrated **IoT edge devices** with cloud-based computer vision pipelines for real-time retail data collection and analytics.

AI Engineer – Government of Bangladesh Project (Contractual) Feb 2024 – Jul 2024

- Led a three-person team to build a multilingual (Bangla/English) intelligent document processing system combining **NLP** and **OCR** for keyword and semantic (context-based) retrieval across PDF, Word, Excel and image formats.
- Delivered the system as an on-premise desktop app (PyQt with a local **ChromaDB** vector store) operating fully offline within a secure government environment, meeting confidentiality and data-governance requirements.

Technical Skills

- Programming:** Python, C/C++, SQL
- AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy
- Computer Vision:** YOLO, ByteTrack, MoveNet, MMAction2, Object Detection, Multi-Object Tracking, Pose Estimation, Action Recognition, OCR
- Generative AI & LLMs:** RAG, LangChain, Hugging Face, Prompt Engineering, Embeddings, Semantic Search, GPT-4o, Stable Diffusion
- Databases:** Pinecone, ChromaDB, MongoDB, Azure SQL
- Cloud & Deployment:** Microsoft Azure, Azure Databricks, Flask, Docker, Git, Linux, CI/CD, Power BI

Key Projects

Sports NFT Generation System

- Developed and evaluated a real-time video analysis pipeline using a fine-tuned object detection model (92% mAP) for automated event segmentation and NFT-ready highlight generation.

Context Searcher

- Developed a multilingual (Bangla/English) RAG-based desktop application for semantic search across documents and images using OCR, LangChain, Embeddings and ChromaDB. Led a three-member team.

Agift Chatbot

- Built a RAG-based chatbot capable of retrieving and generating accurate, context-aware responses using LangChain, vector embeddings, a vector database and OpenAI models.

Universal NFT Generation

- Extended the Sports NFT system to support user-generated videos across festivals, concerts, and other events. Integrated a mobile application and fine-tuned an MMAction2 model for automated NFT clip generation.

Real-Time In-Store Behavior Analytics System

- Designed and developed a real-time customer behavior analytics framework using pose estimation, spatial clustering, and parallel inference for low-latency processing.

Education

Bachelor of Science, Computer Science & Engineering GPA: 3.4 / 4.0 Jan 2019 – May 2024
Comilla University, Bangladesh

Honours & Certifications

- Certified, *Skill Development for Mobile Game & Application Development* – ICT Division, Government of Bangladesh (2022)
- Participant, Inter-University Programming Contest, Comilla University (2019–2021)

References

Hossain Md Erfan – Project Manager, Othello Financial Services Co. Ltd., Osaka, Japan erfan@othello-fs.co.jp
Partha Chakraborty – Associate Professor & Research Supervisor, Comilla University partha.chak@cou.ac.bd